Question 4

$$\tilde{y}_i = b_0 + b_1 \tilde{x}_i$$

$$\frac{1}{n}\sum \tilde{x}_i = 0, \quad \frac{1}{n}\sum \tilde{y}_i = 0$$

1) Ridge adds penalty αb_1^2 to MSE

$$J(b_0, b_1) = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_0 - b_1 \tilde{x}_i)^2 + \alpha b_1^2$$

2) $$\frac{\partial J}{\partial b_0} = -\frac{2}{n}\sum(\tilde{y}_i - b_0 - b_1 \tilde{x}_i) = 0$$

$$\frac{1}{n}\sum(\tilde{y}_i - b_0 - b_1 \tilde{x}_i) = 0$$

$$\frac{1}{n}\sum \tilde{y}_i - b_0 - b_1 \frac{1}{n}\sum \tilde{x}_i = 0$$

$$\sum \tilde{y}_i = 0$$

$$\sum \tilde{x}_i = 0$$

$$b_0 = 0$$

$\longrightarrow$

$$\frac{\partial J}{\partial b_1} = -\frac{2}{n}\sum \tilde{x}_i(\tilde{y}_i - b_1\tilde{x}_i) + 2\alpha b_1 = 0$$

$$-\frac{2}{n}\left(\sum \tilde{x}_i y_i - b_1 \sum \tilde{x}_i^2\right) + 2\alpha b_1 = 0$$

$$-\frac{2}{n}\sum \tilde{x}_i\tilde{y}_i + \frac{2}{n} b_1 \sum \tilde{x}_i^2 + 2\alpha b_1 = 0$$

$$-\frac{1}{n}\sum \tilde{x}_i\tilde{y}_i + b_1\left(\frac{1}{n}\sum \tilde{x}_i^2 + \alpha\right) = 0$$

$$b_1 = \frac{\frac{1}{n}\sum \tilde{x}_i\tilde{y}_i}{\frac{1}{n}\sum \tilde{x}_i^2 + \alpha}$$

3) $$b_1 = \frac{\frac{1}{n}\sum \tilde{x}_i\tilde{y}_i}{\frac{1}{n}\sum \tilde{x}_i^2 + \alpha}$$

As α increases, the denominator increases so, b_1 approaches 0 as α approaches infinity. Increasing α shrinks the slope toward 0 but never exactly reaching 0 unless the numerator is 0.

$\longrightarrow$

4) $J(b_1) = \frac{1}{n}\sum(\tilde{y}_i - b_1\tilde{x}_i)^2 + \alpha|b_1|$

The challenge is that $|b_1|$ is not differentiable at $b_1 = 0$, so we cannot take derivatives everywhere.

1) If $b_1 > 0$ then $|b_1| = b_1$

$J(b_1) = \frac{1}{n}\sum(\tilde{y}_i - b_1\tilde{x}_i)^2 + \alpha b_1$

$-\frac{2}{n}\sum\tilde{x}_i(\tilde{y}_i - b_1\tilde{x}_i) + \alpha = 0$

2) If $b_1 < 0$ then $|b_1| = -b_1$

$J(b_1) = \frac{1}{n}\sum(\tilde{y}_i - b_1\tilde{x}_i)^2 - \alpha b_1$

$-\frac{2}{n}\sum\tilde{x}_i(\tilde{y}_i - b_1\tilde{x}_i) - \alpha = 0$

3) If $b_1 = 0$

At $b_1 = 0$, the function is not differentiable so we use the subgradient

The derivative of the MSE part at $b_1 = 0$ is $-\frac{2}{n}\sum\tilde{x}_i\tilde{y}_i$. In order for $b_1 = 0$ to be optimal, this must be within $[-\alpha, \alpha]$.

So, $\left|-\frac{2}{n}\sum\tilde{x}_i\tilde{y}_i\right| \leq \alpha \rightarrow \left|\frac{1}{n}\sum_{i=1}^{n}\tilde{x}_i\tilde{y}_i\right| \leq \frac{\alpha}{2}$

It is optimal to set $b_1 = 0$ when

$\left|\frac{1}{n}\sum\tilde{x}_i\tilde{y}_i\right| \leq \frac{\alpha}{2}$